

National University of Sciences and Technology (NUST)

Military College of Signals

Assignment 01

Internet of Things (IoT)

Course Code: IT863 • Fall 2026

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Programme: MSSE 32

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Note: Respected Sir, kindly visit my GitHub repository and the live simulation on Streamlit.

GitHub: https://github.com/ali-Hamza817/IoT-Assignment-01

Live simulation: https://iot-assignment-01-5dfaux8t73d8cxdkzp3d5h.streamlit.app/?q=1

Question 1: IoT Sensing Network over 1 km × 1 km

(a) Placement, spacing and number of nodes

- Square grid with node spacing **d = 100 m**.
- First node N1 at **(50, 50)**.
- This gives 10 × 10 = 100 grid positions. The last top-right position **(950, 950)** is the **AP (Access Point)**.
- Total IoT nodes required = **99** (N1 to N99, numbered row by row from the bottom-left).

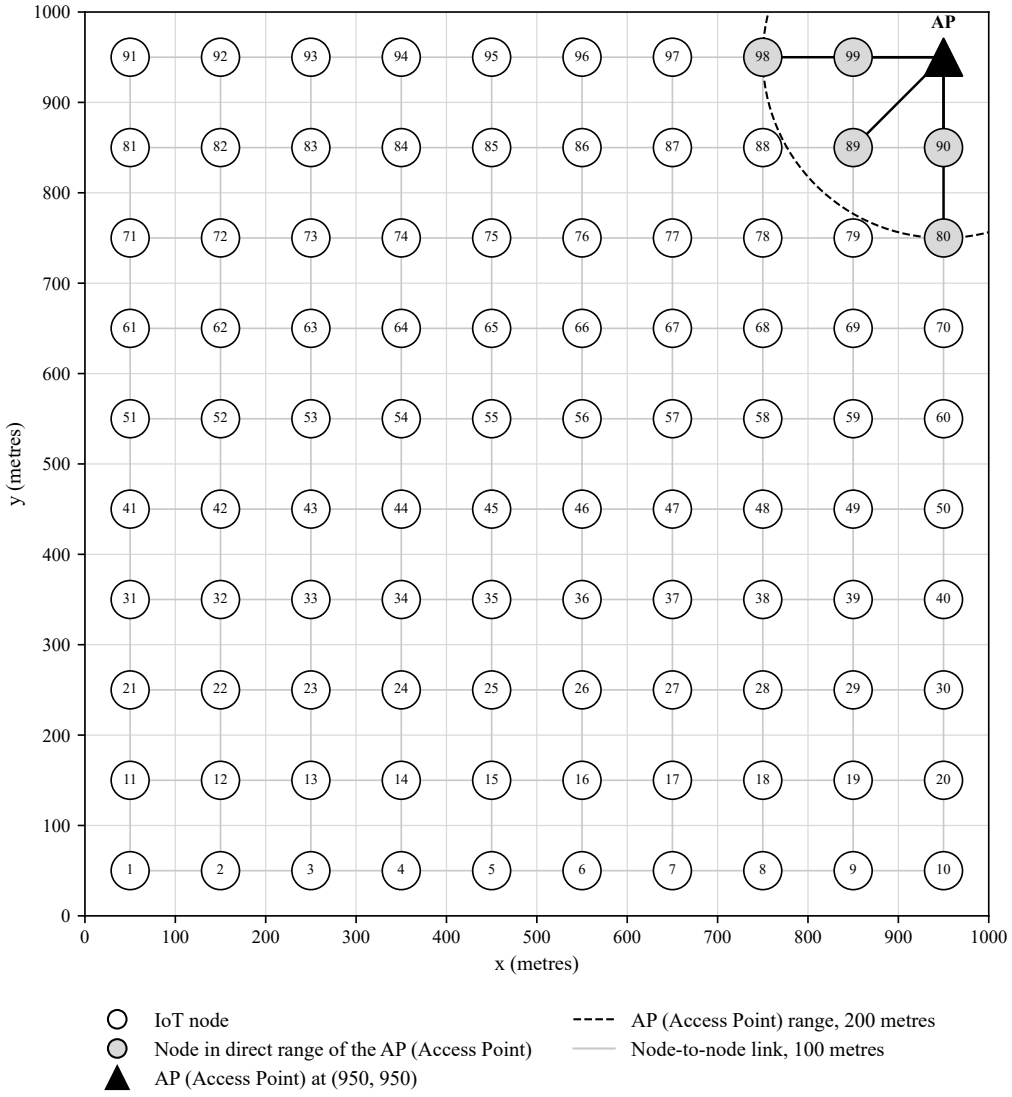


Figure 1: Proposed placement: 99 IoT nodes at 100 m spacing and the AP (Access Point) at (950, 950). Grey nodes are within the 200 m range of the AP (Access Point).

(b) Justification and connectivity

- **Spacing:** neighbouring nodes (left, right, up, down) are exactly 100 m apart, which equals the node range, so every neighbour link satisfies the $\leq$ 100 m constraint.
- **Fewest nodes:** any spacing larger than 100 m breaks the links; any smaller spacing adds nodes that are not needed. 100 m is therefore the largest valid spacing and needs the fewest nodes.
- **Coverage:** the 50 m offset from the edges places each node at the centre of its own 100 m $\times$ 100 m cell, so no point of the area is more than 70.7 m from a node.
- **Every node reaches the AP (Access Point):** from any node, the neighbour 100 m to its right or 100 m above it is one hop closer to the AP (Access Point). Repeating this step always reaches one of the five nodes inside the 200 m range of the AP (Access Point), which then sends directly to the AP (Access Point). The farthest node, N1, needs 17 transmissions; all 99 nodes are connected.

(c) Direct nodes and shortest path

Nodes in direct range of the AP (Access Point) at (950, 950), distance $\leq$ 200 m:

- N99 at (850, 950): 100 m
- N90 at (950, 850): 100 m
- N89 at (850, 850): 141.4 m
- N98 at (750, 950): 200 m
- N80 at (950, 750): 200 m

Selected node: N1 at (50, 50).

Shortest path: N1 $\rightarrow$ N11 $\rightarrow$ N21 $\rightarrow$ N31 $\rightarrow$ N41 $\rightarrow$ N51 $\rightarrow$ N61 $\rightarrow$ N71 $\rightarrow$ N81 $\rightarrow$ N82 $\rightarrow$ N83 $\rightarrow$ N84 $\rightarrow$ N85 $\rightarrow$ N86 $\rightarrow$ N87 $\rightarrow$ N88 $\rightarrow$ N89 $\rightarrow$ AP (Access Point)

- IoT-to-IoT hops = **16**
- Total wireless transmissions = 16 + 1 (final hop N89 $\rightarrow$ AP (Access Point)) = **17**

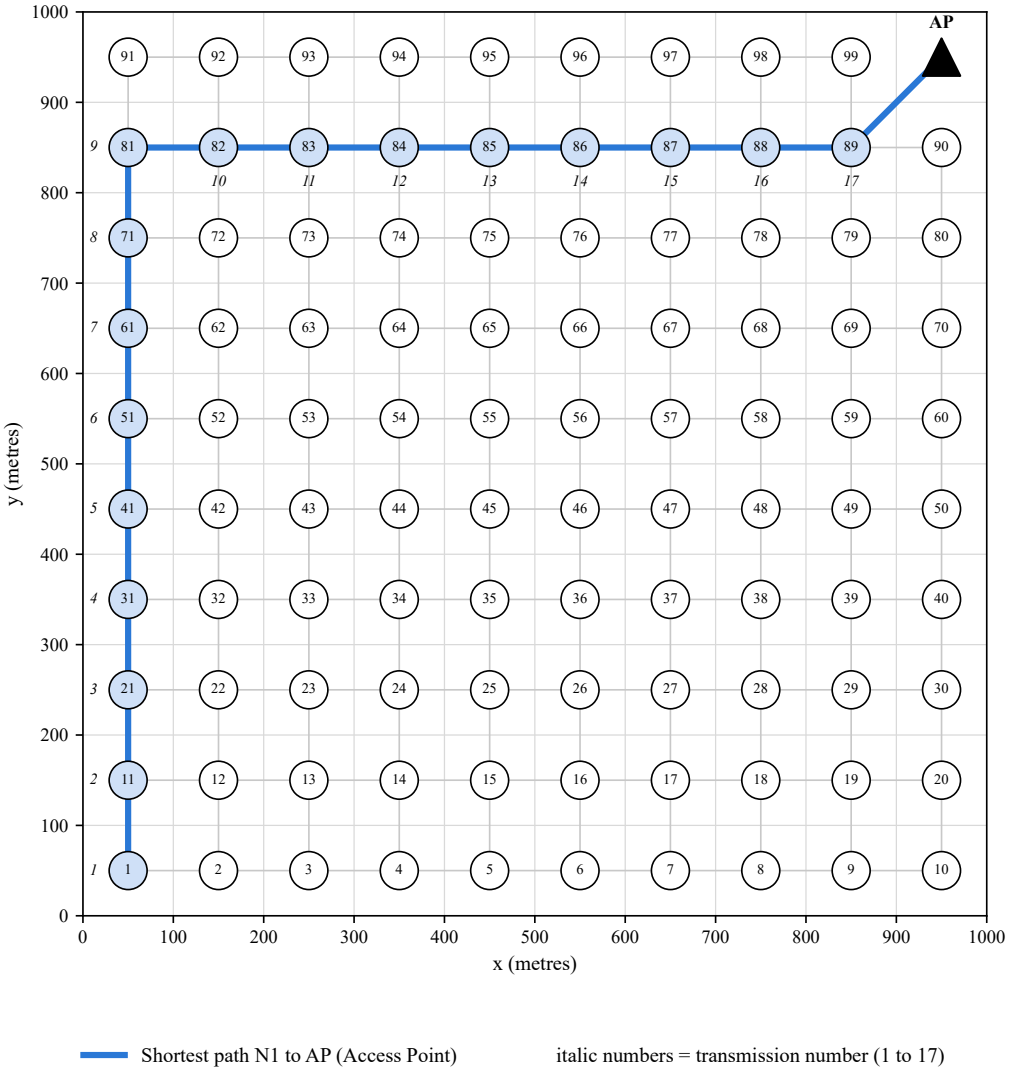


Figure 2: Shortest multi-hop path from N1 to the AP (Access Point): 16 IoT-to-IoT hops and 17 transmissions.

(d) Traffic accumulation and the energy hole

Each node generates one packet per minute. Nodes closer to the AP (Access Point) forward their own packets as well as packets received from nodes farther away.

The forwarding load increases toward the AP (Access Point) because traffic from multiple branches converges near the top-right corner. Nodes adjacent to the AP (Access Point) therefore carry a disproportionately high forwarding burden.

This creates the energy-hole problem: heavily loaded relay nodes deplete their batteries faster than nodes farther away. When these relays fail, distant nodes may lose their routes to the AP (Access Point), reducing network coverage and lifetime.



Figure 3: Forwarding load of every node (node number on top, packets transmitted per minute below). Arrows show the traffic converging on the AP (Access Point).

(e) Technique: unequal clustering

Unequal clustering can reduce the energy burden on nodes near the AP (Access Point). Smaller clusters can be formed near the AP (Access Point), reducing the number of local packets that heavily loaded nodes must process and forward. Cluster heads can aggregate readings before forwarding them.

This reduces redundant transmissions and balances energy consumption, potentially extending the network's operational lifetime.

Question 2: IEEE 802.11 RTS/CTS (Request to Send / Clear to Send) Mechanism

(a) Why can a collision occur?

Sensors A and B cannot hear each other, but both can communicate with the AP (Access Point). If both transmit simultaneously, their signals may overlap at the AP (Access Point), causing a packet collision. This is an example of the hidden-terminal problem: A and B cannot detect each other's transmissions, even though both share the same receiver.

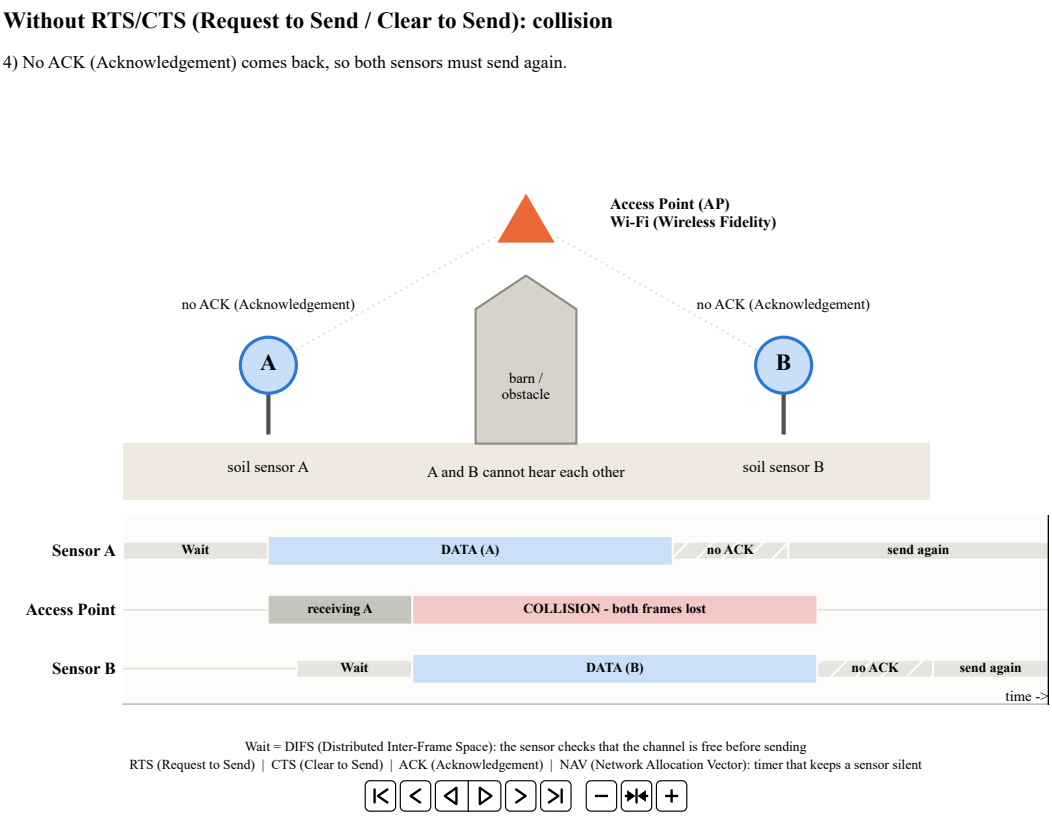


Figure 4: Simulation video: hidden-terminal collision without RTS/CTS (Request to Send / Clear to Send)

Video file attached to this PDF: q2_hidden.mp4 | Watch on GitHub

(b) Complete RTS/CTS (Request to Send / Clear to Send) communication sequence

When sensor A wants to send a soil-moisture reading, the sequence is:

1. **Channel check**
   Sensor A senses the wireless channel and waits until it is free.

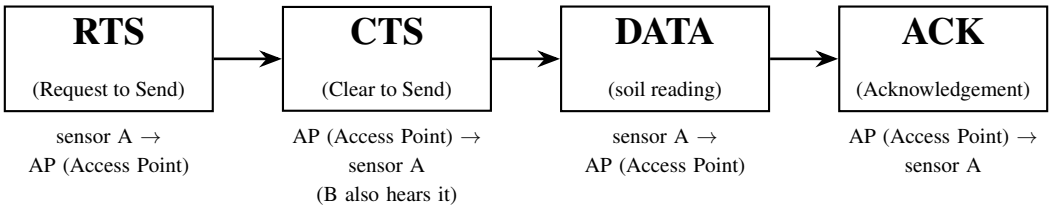
2. **RTS (Request to Send)**
   A sends an RTS (Request to Send) frame to the AP (Access Point), requesting permission to transmit and indicating the expected reservation duration.

3. **CTS (Clear to Send)**
   After receiving the RTS (Request to Send), the AP (Access Point) responds with a CTS (Clear to Send) frame, granting permission and announcing the remaining reserved transmission duration.

4. **DATA transmission**
   A sends its soil-moisture data to the AP (Access Point).

5. **ACK (Acknowledgement)**
   The AP (Access Point) sends an ACK (Acknowledgement) after successfully receiving the data frame.

The basic exchange is:



Short interframe spaces (SIFS) separate the CTS (Clear to Send), DATA, and ACK (Acknowledgement) exchanges as specified by IEEE 802.11.

With RTS/CTS (Request to Send / Clear to Send): no collision

7) The AP (Access Point) replies CTS (Clear to Send) to B; this time A hears it and waits.

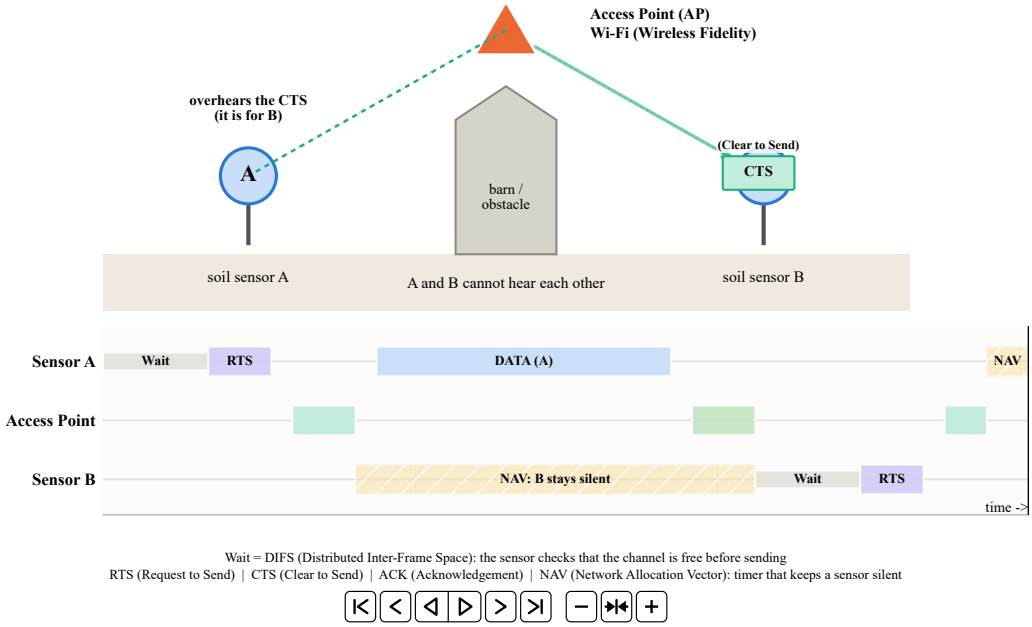


Figure 5: Simulation video: complete RTS (Request to Send) / CTS (Clear to Send) / DATA / ACK (Acknowledgement) exchange

Video file attached to this PDF: q2_rtscts.mp4 | Watch on GitHub

(c) How does CTS (Clear to Send) prevent sensor B from transmitting?

Sensor B receives the AP's (Access Point's) CTS (Clear to Send) frame even though it cannot hear sensor A. The CTS (Clear to Send) contains a duration field that informs B how long the medium is reserved. B updates its Network Allocation Vector (NAV) and defers transmission during the reserved period.

Consequently, B avoids transmitting while A communicates with the AP (Access Point), reducing the likelihood of a collision.

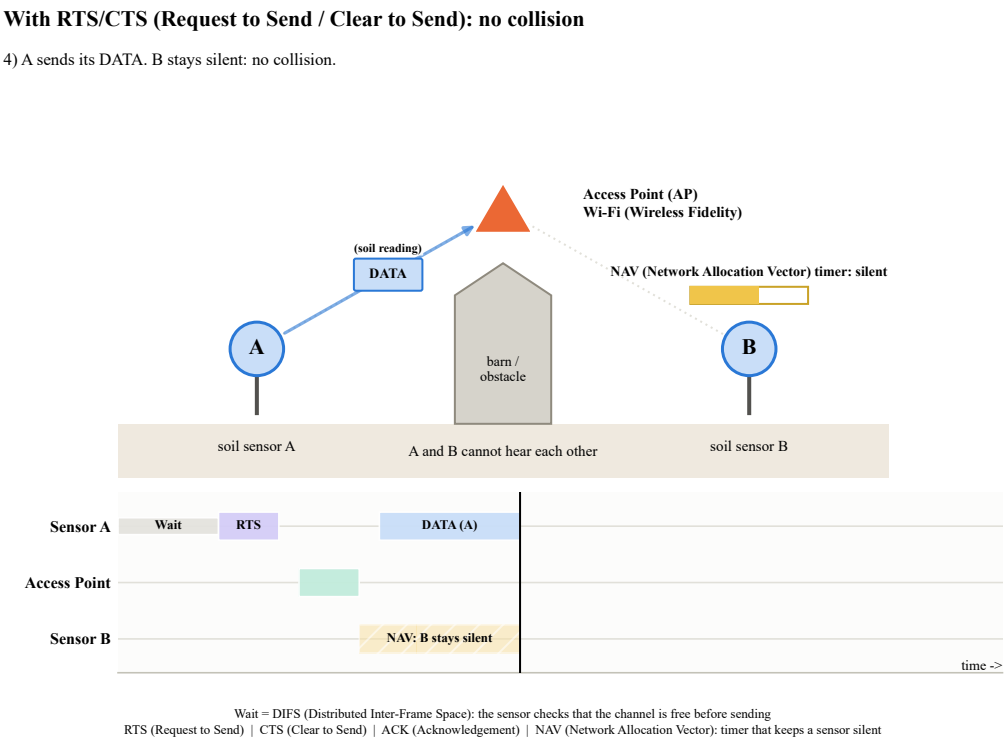


Figure 6: frame from the video above: B's NAV (Network Allocation Vector) keeps B silent while A sends DATA

(d) One benefit and one drawback for battery-powered IoT nodes

Benefit: RTS/CTS (Request to Send / Clear to Send) reduces hidden-terminal collisions, avoiding wasted energy from unsuccessful data transmissions and retransmissions.

Drawback: RTS (Request to Send) and CTS (Clear to Send) add control-frame overhead, extra channel access time, and additional energy consumption. For small, frequent sensor packets, this overhead may outweigh the benefits.

Question 3: Forest-Fire Monitoring Wireless Sensor Network

(a) Star/single-hop versus multi-hop mesh topology

A multi-hop mesh topology is more suitable for the forest-fire monitoring system.

- **Energy consumption:** Distant sensors can forward data through nearby nodes instead of transmitting directly over long distances, potentially reducing transmission energy.
- **Coverage:** Multi-hop communication extends connectivity across a large forest, including locations outside the base station's direct communication range.
- **Reliability:** Multiple possible routes can provide alternative paths when individual nodes or links fail.

A single-hop star topology would require every sensor to reach the base station directly, potentially consuming more energy and limiting coverage.

(b) Flat versus hierarchical/clustered WSN (Wireless Sensor Network) architecture

A hierarchical/clustered architecture is more suitable for 1,000 randomly deployed sensors.

The network is divided into clusters, each managed by a cluster head. Ordinary sensor nodes send their readings to their cluster head, which aggregates the data and forwards it toward the base station.

Advantages include:

- Reduced redundant transmissions.
- Better scalability for a large sensor network.
- Lower communication overhead for ordinary nodes.
- More efficient coordination and energy management.

Cluster heads should be rotated or selected using residual energy and communication cost to avoid premature battery depletion.

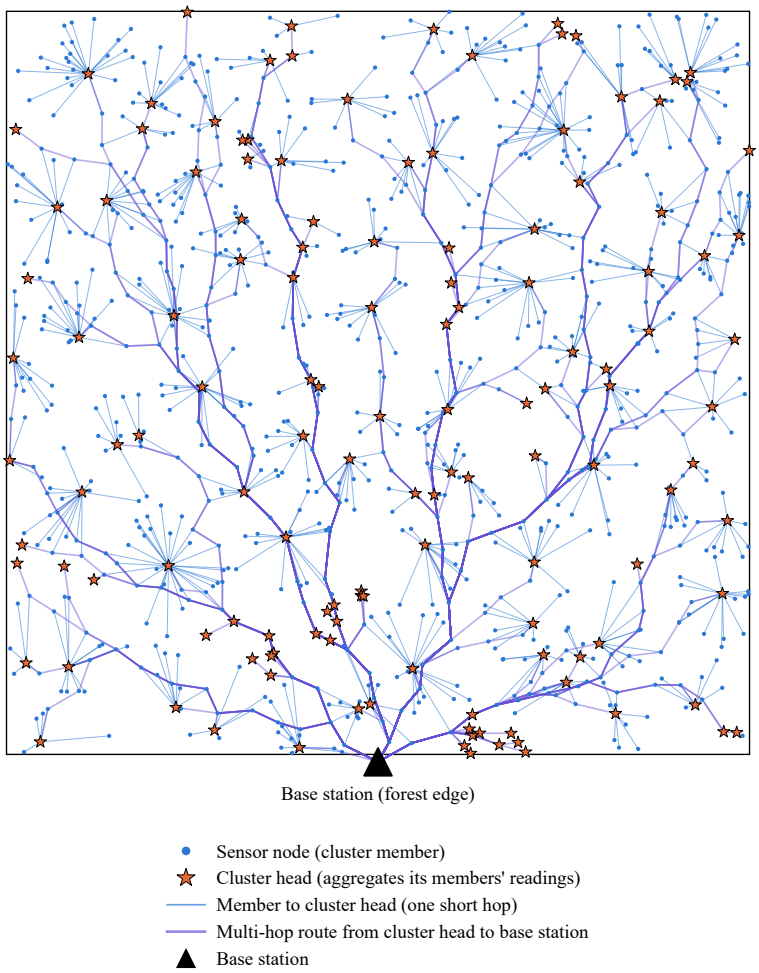


Figure 7: Hierarchical/clustered multi-hop WSN (Wireless Sensor Network) for the forest: 1,000 sensor nodes over a 2 kilometre $\times$ 2 kilometre forest. Each cluster head aggregates its members' readings and forwards them to the base station over a multi-hop route.

(c) How data aggregation extends network lifetime

Data aggregation combines readings from multiple sensors into a compact summary before forwarding them. For example, a cluster head can calculate average temperature and humidity while preserving maximum temperature and smoke readings that may indicate a fire.

Without aggregation, many individual packets must travel toward the base station. With aggregation, fewer packets are transmitted over long distances, reducing communication energy and congestion. This helps conserve battery power and extends network lifetime while retaining important fire-detection information.

Declaration: AI tools were used for the simulations and the coding; all of this work has been reviewed and edited by me.